PF LAB NO. 4

Name: Hamza Hasnain
Roll No: 48
Section: BS CS A
PF Lab No: 4

Written Work — Task 1 — Positive number

If the entered number is negative, print its positive equivalent; otherwise print the entered number as requested.

Code / Answer (write on A4 paper):

```
#include <iostream>
using namespace std;
int main() {
double n;
cout << "Enter a number: ";
cin >> n;
if (n < 0)
cout << "Positive number = " << -n << endl;
else
cout << "Number entered = " << n << endl;
return 0;
}
```

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Written Work — Task 2 — Positive, negative or zero

Use if, else if and else to classify an integer.

Code / Answer (write on A4 paper):

```
#include <iostream>
using namespace std;
int main() {
int n;
cout << "Enter an integer: ";
cin >> n;
if (n > 0) cout << "Positive\n";
else if (n < 0) cout << "Negative\n";
else cout << "Zero\n";
return 0;
}
```

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Written Work — Task 3 — Grade from marks

Apply the thresholds in the lab sheet: A >= 75, B >= 60, C >= 45, otherwise D.

Code / Answer (write on A4 paper):

```
#include <iostream>
using namespace std;
int main() {
int mark;
cout << "Enter marks: ";
cin >> mark;
if (mark >= 75) cout << "Grade A\n";
else if (mark >= 60) cout << "Grade B\n";
else if (mark >= 45) cout << "Grade C\n";
else cout << "Grade D\n";
return 0;
}
```

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Written Work — Task 4 — Largest number and digit count

Find the largest of three numbers using nested if-else, then check whether a number is positive and count its digits.

Code / Answer (write on A4 paper):

```
#include <iostream>
using namespace std;
int main() {
int a, b, c, n, digits = 0;
cout << "Enter three numbers: ";
cin >> a >> b >> c;
int largest;
if (a > b) {
if (a > c) largest = a;
else largest = c;
} else {
if (b > c) largest = b;
else largest = c;
}
cout << "Largest = " << largest << endl;
cout << "Enter a positive number: ";
cin >> n;
if (n > 0) {
do { digits++; n /= 10; } while (n > 0);
cout << "Digits = " << digits << endl;
} else {
cout << "Number is not positive.\n";
}
return 0;
}
```

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Laptop — Code Editor: Task 1 — Positive number

Code::Blocks / C++ source

```
#include <iostream>
using namespace std;
int main() {
double n;
cout << "Enter a number: ";
cin >> n;
if (n < 0)
cout << "Positive number = " << -n << endl;
else
cout << "Number entered = " << n << endl;
return 0;
}
```

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Laptop — Program Output: Task 1 — Positive number

```
Enter a number: -8
Positive number = 8
```

Illustrative terminal output — run the code on your laptop to capture the actual result.

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Laptop — Code Editor: Task 2 — Positive, negative or zero

Code::Blocks / C++ source

```
#include <iostream>
using namespace std;
int main() {
int n;
cout << "Enter an integer: ";
cin >> n;
if (n > 0) cout << "Positive\n";
else if (n < 0) cout << "Negative\n";
else cout << "Zero\n";
return 0;
}
```

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Laptop — Program Output: Task 2 — Positive, negative or zero

```
Enter an integer: 5
Positive
```

Illustrative terminal output — run the code on your laptop to capture the actual result.

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Laptop — Code Editor: Task 3 — Grade from marks

Code::Blocks / C++ source

```
#include <iostream>
using namespace std;
int main() {
int mark;
cout << "Enter marks: ";
cin >> mark;
if (mark >= 75) cout << "Grade A\n";
else if (mark >= 60) cout << "Grade B\n";
else if (mark >= 45) cout << "Grade C\n";
else cout << "Grade D\n";
return 0;
}
```

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Laptop — Program Output: Task 3 — Grade from marks

```
Enter marks: 82
Grade A
```

Illustrative terminal output — run the code on your laptop to capture the actual result.

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Laptop — Code Editor: Task 4 — Largest number and digit count

Code::Blocks / C++ source

```
#include <iostream>
using namespace std;
int main() {
int a, b, c, n, digits = 0;
cout << "Enter three numbers: ";
cin >> a >> b >> c;
int largest;
if (a > b) {
if (a > c) largest = a;
else largest = c;
} else {
if (b > c) largest = b;
else largest = c;
}
cout << "Largest = " << largest << endl;
cout << "Enter a positive number: ";
cin >> n;
if (n > 0) {
do { digits++; n /= 10; } while (n > 0);
cout << "Digits = " << digits << endl;
} else {
cout << "Number is not positive.\n";
}
return 0;
}
```

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Laptop — Program Output: Task 4 — Largest number and digit count

```
Enter three numbers: 8 12 5
Largest = 12
Enter a positive number: 456
Digits = 3
```

Illustrative terminal output — run the code on your laptop to capture the actual result.